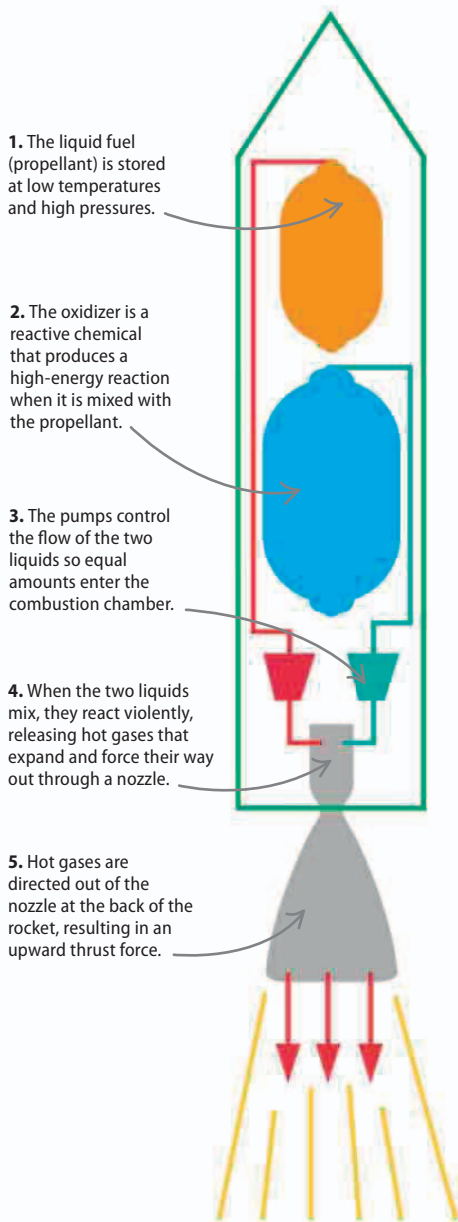


Rocket engine

Rocket engines do not burn their fuels in air. Instead, the fuel is mixed with another chemical, called an oxidizer, which creates a very hot and vigorous reaction. The hot expanding gases produced by the reaction are forced out of a small nozzle. The action of the gas leaving the engines results in a reaction force that drives the rocket forward.

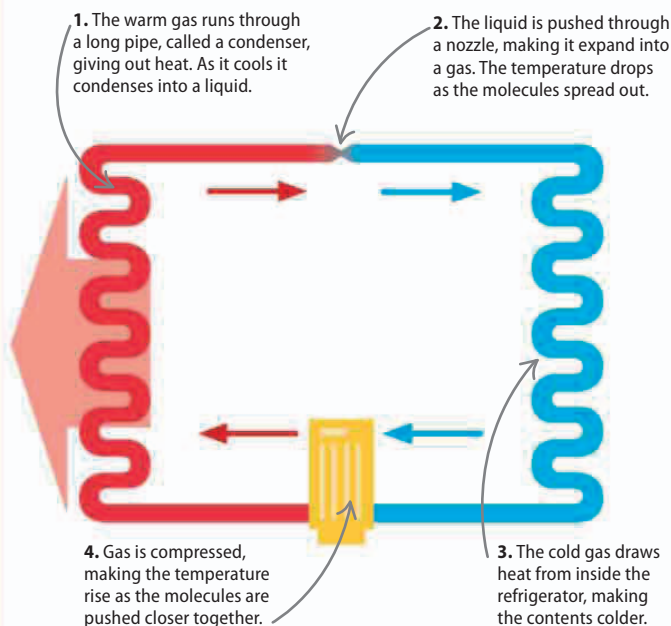
▽ How a liquid-fueled rocket engine works

Unlike a jet engine, a rocket engine carries both the fuel and oxidizer on board. Smaller rockets, such as fireworks, use solid fuels, while the largest rockets have liquid fuels.



Refrigeration

Cold is the absence of heat, and a refrigerator chills food by removing heat from the internal storage space. Heat energy always moves from hot places to cold ones. A refrigerator works by passing a cold gas behind the storage space, so heat from the air inside moves to that gas, making the air colder. The cold gas is produced by expanding a liquid very rapidly. The temperature drops as the molecules spread out, thus preserving food and drinks.



△ Refrigeration cycle

In a refrigerator, a refrigerant (a substance used for cooling) travels around a system of pipes. First, heat radiates from the warm refrigerant. Second, the refrigerant begins to expand and cool. Third, the cold refrigerant cools the refrigerator because thermal energy moves from the refrigerator to the refrigerant. Finally, the compressor squeezes the refrigerant so it gets warmer as it begins to release its thermal energy.

REAL WORLD

Microwave oven

A microwave heats food using high-energy microwaves, which are absorbed by the bonds in water and fat molecules. These vibrate, which causes the food to heat up.

